

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 12

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • General physiological solvent: most biological reactions take place between substances dissolved in water (1) • Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes (1) • High specific heat capacity moderates temperature changes; evaporation absorbs heat (cooling) (1) • Anomalous expansion: ice floats so water bodies do not freeze solid; absorbs UV protecting early marine life; provides aquatic habitats (1)	4
01.2	<i>1 mark for each point.</i> • From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen; at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) (1) • Oxygen in the stratosphere absorbed UV, forming ozone (O₃) in a dynamic equilibrium of O, O₂ and O₃ (1) • The ozone layer reduced UV reaching the surface so organisms could survive out of the water (1) • Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved; many anaerobic archaea died out (1)	4
01.3	<i>1 mark for each point.</i> • Photosynthesis removed CO₂ (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels (1) • This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • All greenhouse gases expressed as CO₂ equivalent (allowing for different warming strengths) (1) • Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis) (1) • A fair share of the infrastructure: roads, car parks, signals, street lighting (1)	3
02.2	<i>1 mark for each point.</i> • Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels (1) • Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time (1)	2
02.3	<i>1 mark for each point.</i> • Hardy, disease-resistant Boran cattle; regular tick dipping reduces ticks for wild herbivores too (1) • Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth (1) • Electric fences keep game in and poachers out; biological corridors let animals (except rhinos) move to other reserves; problem elephants relocated (1) • Monitoring: radio-collared lions, drones, camera traps, track surveys; endangered species (Grevy's zebra, northern white rhino) protected (1) • Community links: ecotourism income funds health, education and water projects; local employment; cattle breeding help for local farmers (1)	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2.	
03.1	<i>1 mark for each point.</i> • Overall decline from 62% to 22% (1) • Sharp falls in 1998 (to 35%) and 2016 (to 18%) coinciding with bleaching events, with partial recovery after each (1)	2
03.2	<i>1 mark for each point.</i> • The polyps expel their symbiotic zooxanthellae so the coral loses colour; bleached corals are alive but grow very slowly and may die if algae do not return (1) • Raised water temperature (climate change / El Niño) (1) • Pollution: sewage, pesticides, cyanide, sunscreen; low oxygen from zooplankton blooms after overfishing (1) • High light / solar activity; ocean acidification hinders skeleton formation (1)	4
03.3	<i>1 mark for each point.</i> • Coastal construction and dredging increase sedimentation / turbidity covering and killing polyps (1) • Sewage and fertiliser nutrients cause algal growth that shades corals and prevents egg release (1) • Physical damage from anchors, divers, souvenir collection; loss of mangroves / seagrass nurseries that trap sediment (1)	3
03.4	<i>1 mark.</i> • Protected breeding populations grow larger and produce surplus young / adults that spill over into fished areas (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 60-63; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Other MEDCs (USA, Canada, Australia) also had surpluses; Japan already had suppliers (1) • Eastern Europe / USSR needed food but could not pay (1) • Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production (1)	3
04.2	<i>1 mark for each point.</i> • Quotas limiting production (eg milk) (1) • Set-aside payments for taking land out of production (1) • Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees) (1) • Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming (1)	4
04.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape (1) • Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat (1) • Beetle banks of tussocky grass - natural pest predators (1) • Skylark nesting plots; wild bird seed / nectar plants - winter food and pollinators (1) • In-field tree protection; controlled stocking density to reduce compaction; ditch management; dry-stone walls; woodland management (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.	
05.1	<i>1 mark for each point.</i> • Ship rudders that turn in tighter circles / aerogenerator blades that start turning at lower wind speeds (1) • Water-repelling surface of lotus flowers (1) • Coatings for aircraft and ships that reduce fuel consumption / hospital coatings that bacteria cannot stick to (1) • Hooked burrs on seeds that stick to animal fur (1)	4
05.2	<i>1 mark for each point.</i> • The mound absorbs sunlight and warms; hot air inside rises, creating a convection current that draws stale air out of the lower parts (solar-driven ventilation) (1) • Air blowing over the mound has a lower pressure which also draws air out (1) • Copied in shopping complexes / office blocks to give natural ventilation and cooling without air conditioning, so less energy / electricity is used (1)	3
05.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Marsupials: young develop in the pouch so are easier to study than a human foetus, helping understanding of developmental problems (1) • Hippopotamus skin secretes hipposudoric acid (natural sunscreen and antimicrobial) which may improve treatment of burns (1) • Marine sponges produce proteins that prevent rejection of grafts, being developed to prevent rejection of transplanted organs (1)	2
05.4	<i>1 mark.</i> • They are light but strong: hollow with internal struts that prevent bending / breaking, copied in lightweight bridges and roofs (1)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11-14; AQA spec 3.1.2.1.1.	
06.1	<i>Correct answer only.</i> • Environmental DNA (eDNA) analysis (1)	1
06.2	<i>1 mark for each point.</i> • Transmitters are too heavy for small birds / their batteries would not last for a long-term study (1) • Geolocator tags are small and only record time and light levels; latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded (1)	2
06.3	<i>1 mark for each point.</i> • Each separate (regional) population has its own unique gene pool (1) • DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
07.1	<i>1 mark for each point.</i> • Hard release: individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) (1) • Soft release: gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds (1) • Problems faced: finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups (1)	3
07.2	<i>Any 5 from the points below - 1 mark each, maximum 5.</i> • Large enough suitable habitat for a viable population (1) • Reliable food supply (1) • Low predation risk / original threat removed (1) • Suitable breeding sites (1) • Water (1) • Support of the local human population (1) • Official / legal protection of habitat and species (1)	5
07.3	<i>1 mark for each point.</i> • Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed (1) • Identifies causes of failure so future releases can be improved / extra support given (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	

Question	Marking guidance	Marks
08.1	<i>Up to 6 marks in total; award marks in proportion to the completeness of each point.</i> • Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance; indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met (up to 2) • Single species: easy planting, thinning and felling over large areas; but each tree species supports its own community so few species overall (up to 2) • Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them (up to 2) • Simple age structure: uniform size makes management easy but gives little variety of light, temperature, wind; no old trees or dead wood for invertebrates and woodpeckers (up to 2) • Mixed woodland: varied niches, stratification, dead wood, indigenous trees each with their own dependent species (up to 2)	6
08.2	<i>1 mark for each point.</i> • Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline (1) • Continuing to grow them ties up land and capital for little extra timber; natural forests keep old and dead trees (1)	2
08.3	<i>1 mark for each point.</i> • Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp (1) • Wood fuel replaced by fossil fuels / electricity; so traditional management declined and woods became uniform (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	
09.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Extensive cereal 5.0; intensive cereal 3.0 (1) • Intensive beef 0.1 (1)	2
09.2	<i>1 mark for each point.</i> • Intensive cereal is most productive per hectare (60 GJ) but less efficient (3.0) than extensive (5.0) (1) • Each extra unit of input gives a smaller extra yield (law of diminishing returns) (1) • Beef has very low productivity and efficiency because energy is lost at an extra trophic level (1)	3
09.3	<i>Up to 3 marks in total; award marks in proportion to the completeness of each point.</i> • Any input that increases productivity but requires the use of energy (usually fossil fuel) (up to 2) • Manufacture of nitrate fertiliser (Haber process) and pesticides; fuel for machinery; pumping irrigation water; grain drying; food processing and transport; embodied energy of machinery (up to 2)	3

Question	Marking guidance	Marks
09.4	1 mark for each point. - Because of diminishing returns, the first units of input on unfertilised land give the largest yield increases (1) - Moving inputs from intensively farmed land (where extra input adds little) to extensively farmed land gives a greater total yield - though land, cost and distribution limit this (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-327; AQA spec 3.5.1.2.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food - Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish - Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores - Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced - Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data - Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management	9
	Model answer - Level 3 (9 marks)	
	Wild catches have not increased since about 1990 because most stocks are fully or over-exploited, while aquaculture has grown rapidly and now approaches the wild catch, so the question is how far it can take over. Its advantages are real: limiting factors, breeding, gender and disease can be controlled, yields per area are high, and fish have low food conversion ratios because they need not support their weight or keep warm; herbivores such as tilapia and carp and filter-feeding molluscs need no fishmeal at all.	

Question	Marking guidance	Marks
	Its limits are equally real. The carnivorous species consumers prefer - salmon, shrimp - are fed fishmeal from wild forage fish, so their farming still depends on fishing and depletes species such as sandeels on which puffins rely; some species, such as bluefin tuna, will not breed in captivity. Farms destroy mangroves, release organic waste, pesticides and antibiotics that breed resistance, spread lice, allow escapes that contaminate wild gene pools and provoke predator control, though siting, closed systems, polyculture and vaccines can reduce these. Meanwhile fishing itself can be sustainable if quotas based on good stock data, gear design, size and effort limits and no-take zones are enforced.	
	Aquaculture can therefore supply most of the growth in demand, especially through herbivorous fish and shellfish, and can relieve pressure on wild stocks, but it cannot fully replace fishing while carnivorous species depend on wild fishmeal and some species cannot be farmed. Both must be managed sustainably; neither is a substitute for the other.	
	Weaker answer - Level 1 (2 marks)	
	Aquaculture is farming fish instead of catching them. Fish farms can produce a lot of fish and they do not run out like wild fish do. But fish farms cause pollution and diseases can spread to wild fish. Salmon farms feed the salmon on fish which have to be caught. So aquaculture can replace some fishing but not all of it.	
	What makes the difference	
	• Level 1 reaches a reasonable conclusion but supports it with two sentences; Level 3 gives the context (flat wild catch since 1990), the advantages in detail and the impacts with ways to reduce them.	
	• Level 3 distinguishes between species types (herbivores and molluscs vs carnivores) - the key to 'to what extent'.	
	• Level 3 also considers whether fishing can be made sustainable, which the comparison requires.	
10.2	<i>1 mark.</i> • A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Question	Marking guidance	Marks
11.1	Discuss how human activities influence biodiversity, using examples from a range of different contexts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Direct exploitation: food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil); deliberate eradication of predators, competitors, vectors, pests • Changes in abiotic factors: water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage; inactivity (stopping plagioclimax management) • Changes in biotic factors: introduced competitors, predators, pathogens, hybridisation; loss of pollinators, seed dispersers, keystone species; food chain impacts (sea otters, puffins, jellyfish) • Habitat destruction: deforestation, farmland expansion, urbanisation, mining, reservoirs; fragmentation and edge effects • Contexts: temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica; agriculture, fishing, forestry • Positive influences: habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	25

Question	Marking guidance	Marks
11.2	Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why monitor: species present, population size, density, distribution, survival, age structure, breeding rates, movements; baseline and evaluation; Red List categorisation and quotas rely on it • Standard techniques and their limitations: quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR; sampling design (random / systematic, size, number, timing, standardisation) • New technologies: satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • How they improve validity: representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • Examples: penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked • Limitations: cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	25